

The University of Western Australia
SCHOOL OF MATHEMATICS & STATISTICS
AMO TRAINING SESSIONS

Australian Mathematics Olympiad, 2009 Trial Problems

1. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x)f(y) - f(xy) = x + y$$

for all $x, y \in \mathbb{R}$.

2. In a latest theory of particle physics there are three fundamental particles. When two particles of different types collide they are replaced by a particle of the third type. Two particles of the same type never collide. Prove that if an experiment begins with an equal number of particles of each type it cannot end with just one particle remaining.
3. A chord PQ with midpoint R is drawn in a circle with diameter AB . Perpendiculars PS and QT are dropped onto AB .
- (i) Prove that $\triangle RST$ is isosceles.
- (ii) Prove that $\triangle RST$ is equilateral if and only if $2PQ = AB$.
4. Find the smallest positive integer which has exactly 30 positive divisors.
5. Let $a, b, c \in \mathbb{R}$ such that $0 < a, b, c \leq 1$. Prove that

$$\frac{a}{bc+1} + \frac{b}{ca+1} + \frac{c}{ab+1} \leq 2.$$

6. Number plates in a certain country have 6 digits ranging from 000000 to 999999. How many number plates can be issued such that any 2 number plates differ in at least 2 places?
7. For any positive integer n , let $T(n)$ be the number of positive divisors of n . Find all n for which the sequence

$$n, T(n), T(T(n)), \dots$$

does not contain a perfect square.

8. A quadrilateral $ABCD$ is inscribed in a circle so that AB is a diameter. Suppose that AB and CD intersect at I , AD and BC intersect at J , AC and BD intersect at K and let N be a point on AB . Prove that $IK \perp JN$ if and only if N is the midpoint of AB .